

MANTHAN

NEET 2027

PHYSICAL CHEMISTRY

CHEMICAL EQUILIBRIUM

LECTURE -02

Subtopic = EQUILIBRIUM STATE



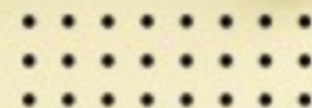
Sarvesh Sir



Today's goal



- ▶ **EQUILIBRIUM STATE**
- ▶ **EQUILIBRIUM CONSTANT**
- ▶

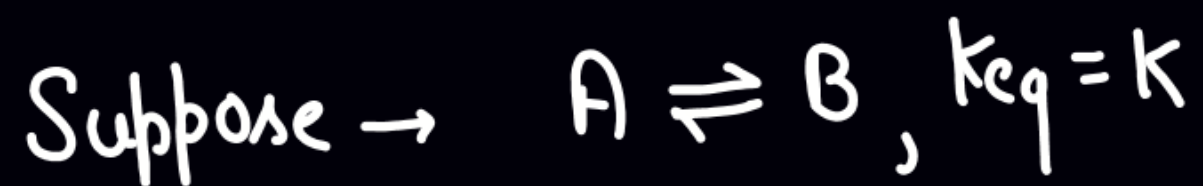


Equilibrium Constant also depend upon Stoichiometric Coefficient of Rxn



Stoichiometric coefficient

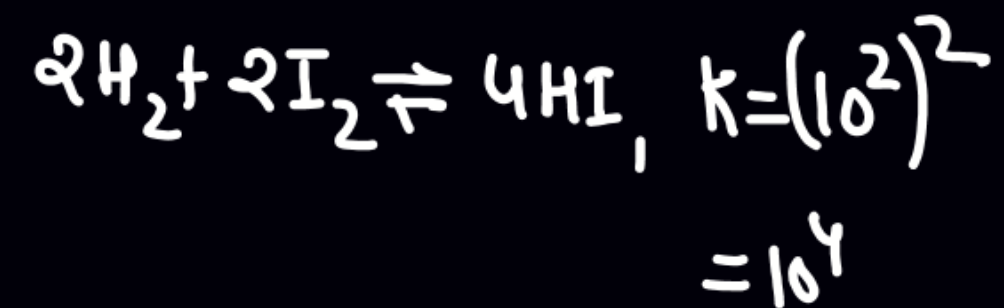
→ Eq. Const depend upon stoichiometric coefficient of rxn.



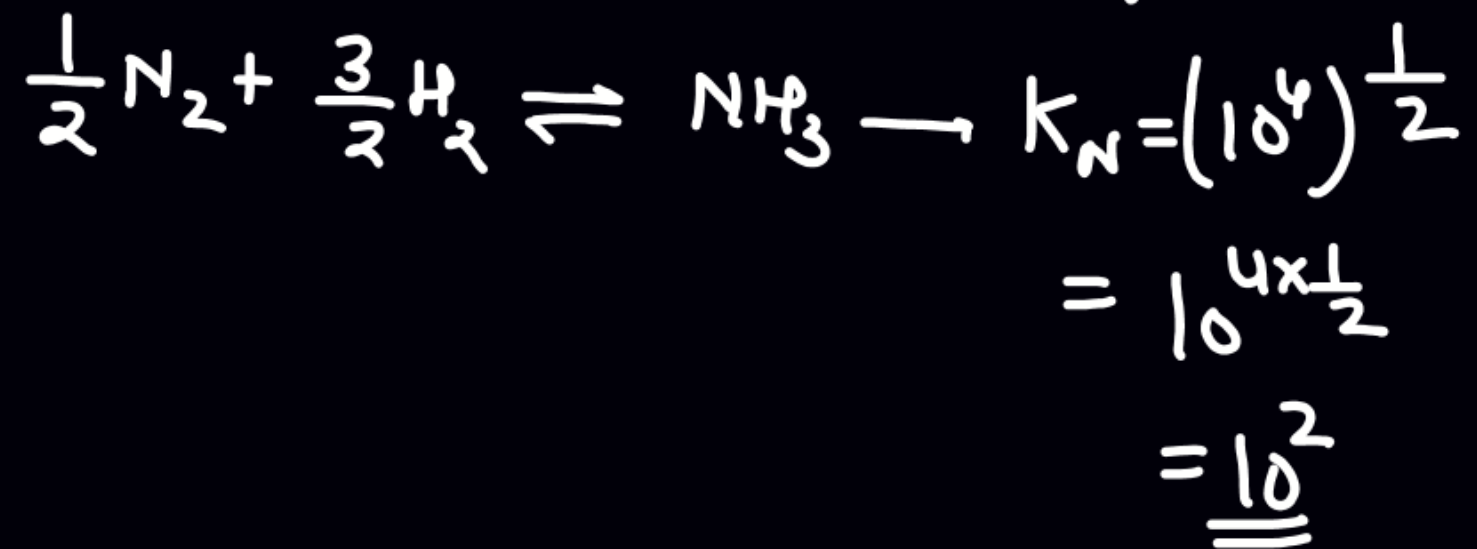
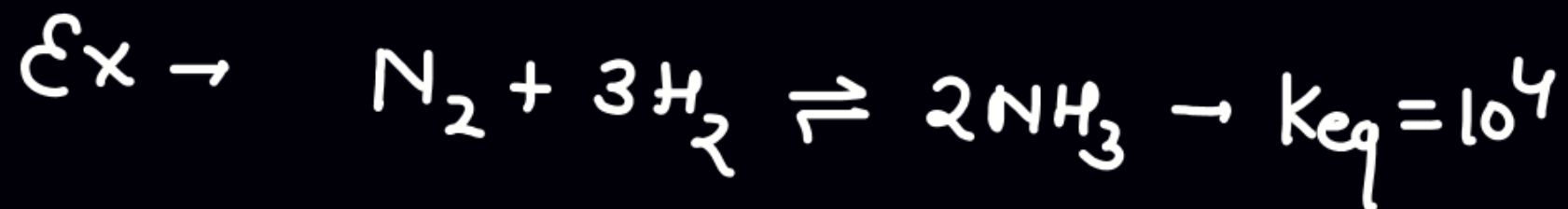
Rule-1 Multiple by 'n'



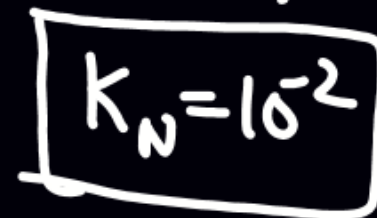
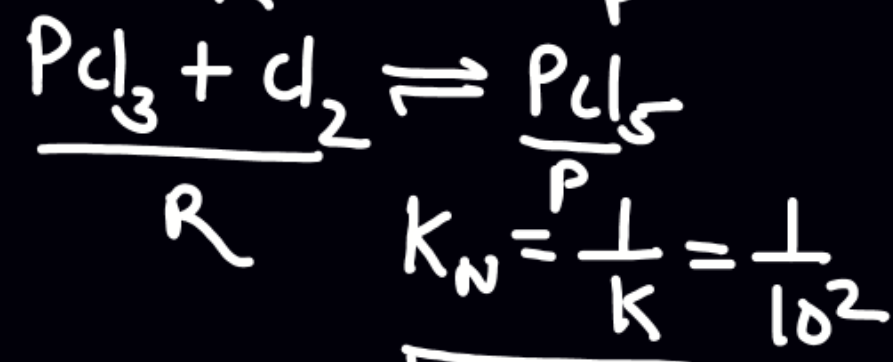
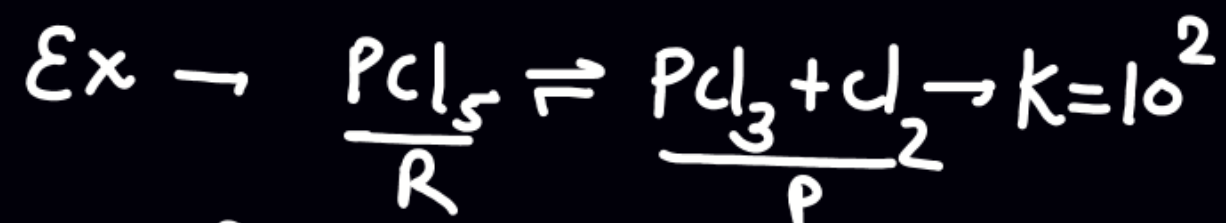
$$K_{n.} = (K)^n$$

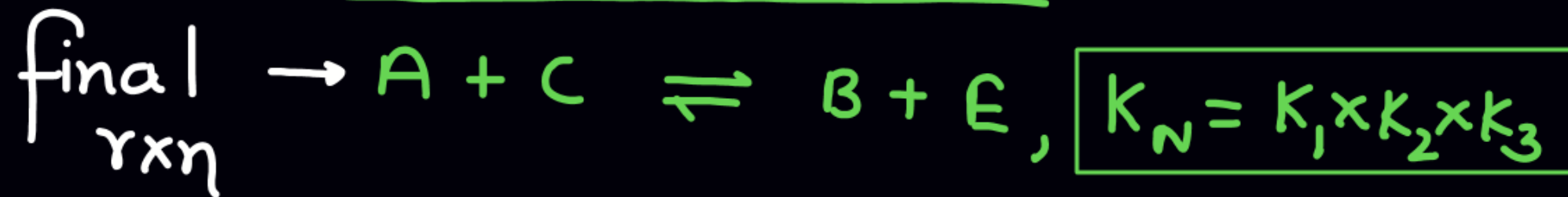
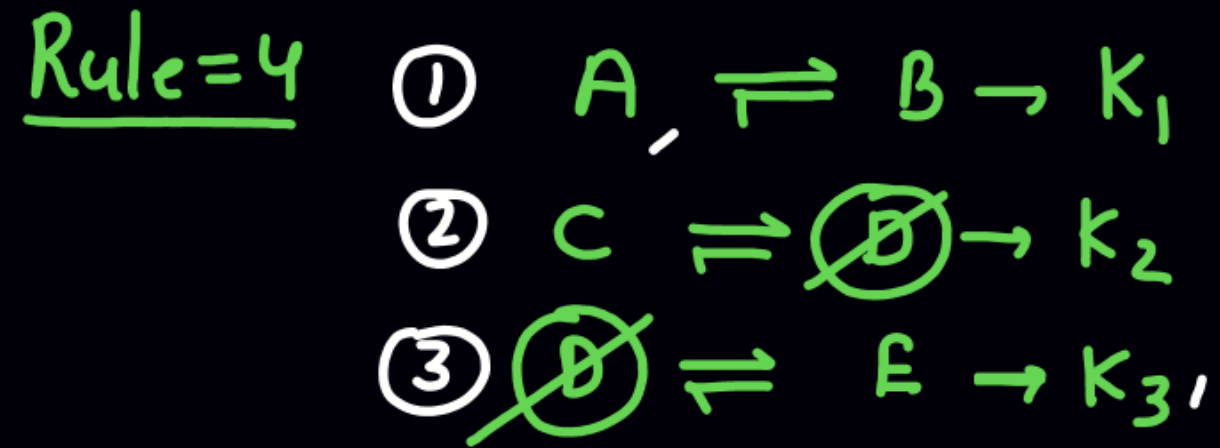


Rule-2 → divide by n or multiply by $\frac{1}{n}$



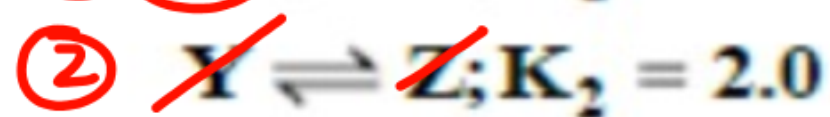
Rule-3 → Opposite Rxn



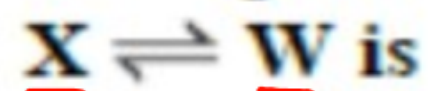


Csd = Equation को math की Equation की तरह solve करेंगे, But Eq. Const को Eq. Const के Rule-1, 2, 3, 4 से solve करेंगे.

15. For the given hypothetical reactions, the equilibrium constants are as follows:



The equilibrium constant for the reaction



[08 April, 2024 (Shift-I)]

(a) 6.0

(b) 12.0

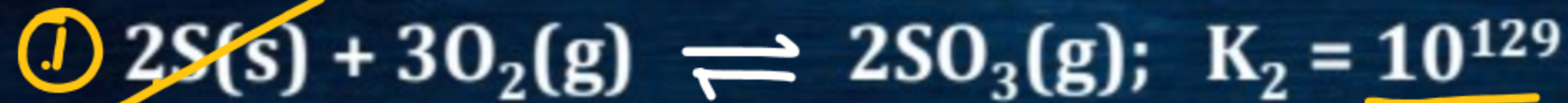
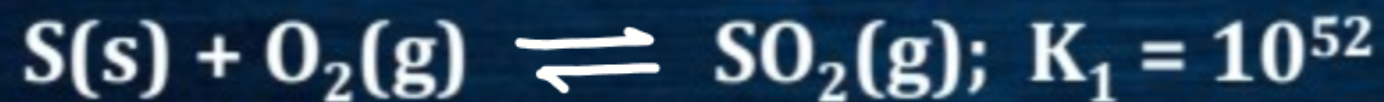
~~(c) 8.0~~

(d) 7.0

① + ② + ③

$X \rightleftharpoons W, \rightarrow K = 1 \times 2 \times 4 = 8$

For the following reactions, equilibrium constants are given:



The equilibrium constant for the reaction: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ is:

[2019 Main, 8 April II]

$$x + 3 = y + 2$$

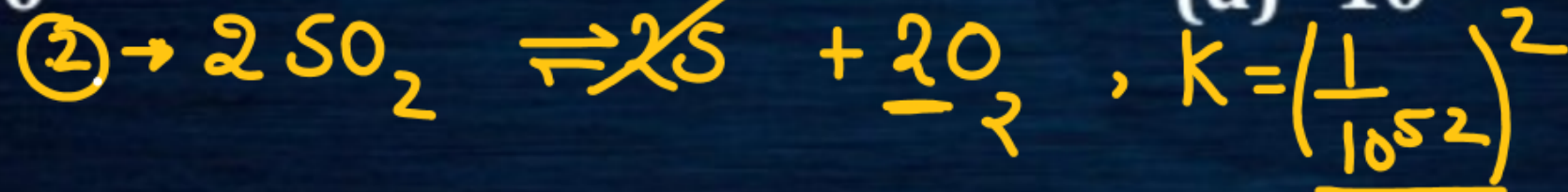
$$x + 3 - 2 = y \Rightarrow x + 1 = y$$

~~(a) 10^{25}~~

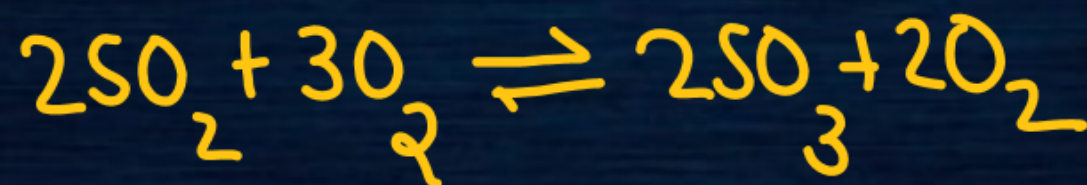
(b) 10^{77}

(c) 10^{154}

(d) 10^{181}



$\textcircled{1} + \textcircled{2}$

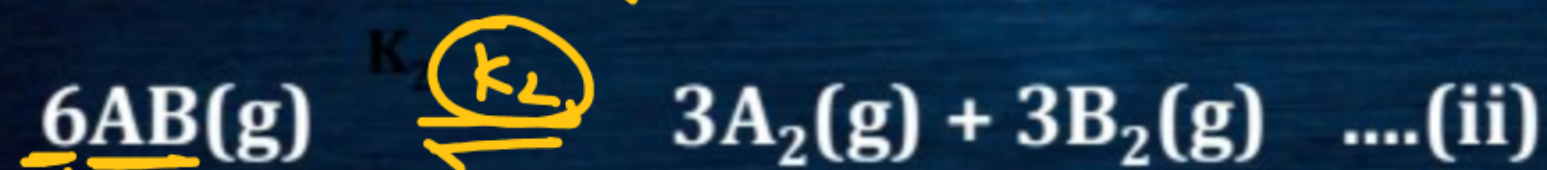


$$2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3, K = (10^{129}) \times \left(\frac{1}{10^{104}}\right) = 10^{129-104} = \underline{\underline{10^{25}}}$$



Consider the following reversible chemical reaction,

[2019 Main, 9 Jan II]



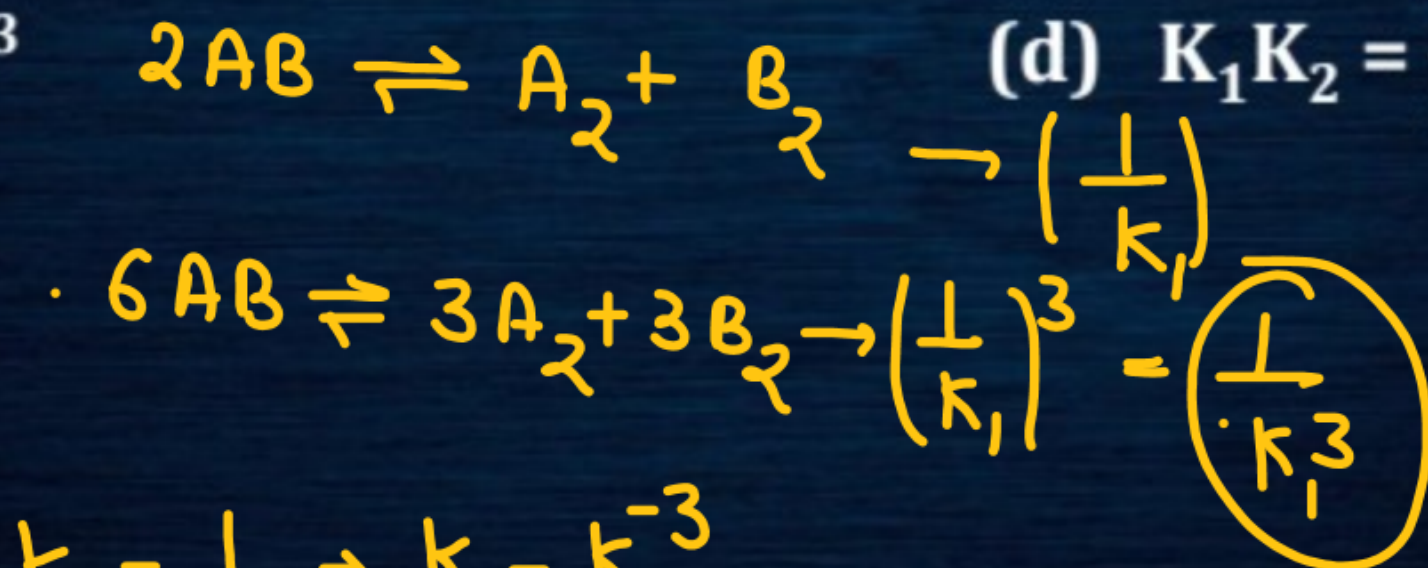
The relation between K_1 and K_2 is:

(a) $K_2 = K_1^3$

(b) $K_1 K_2 = 3$

~~(c) $K_2 = K_1^{-3}$~~

(d) $K_1 K_2 = \frac{1}{3}$



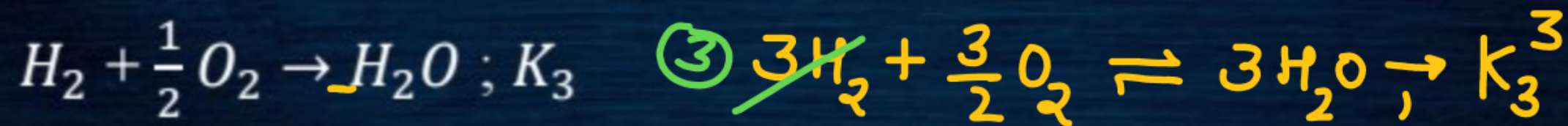
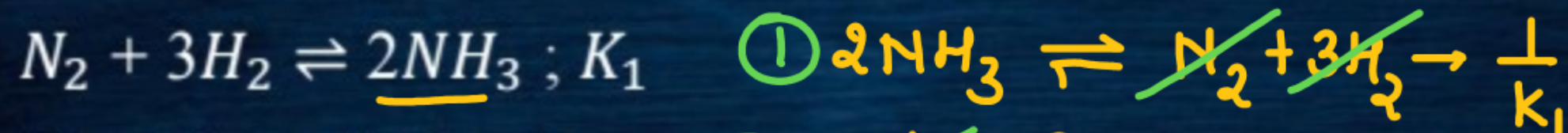
$$K_2 = \frac{1}{K_1^3} \Rightarrow K_2 = K_1^{-3}$$



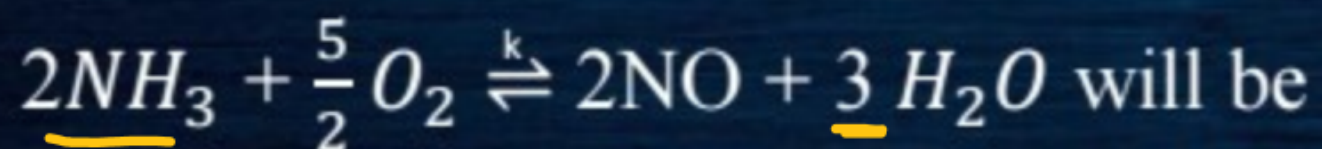
Q1. The equilibrium constants of the following are

$$1 + \frac{3}{2} = \frac{2+3}{2}$$

NEET-2017



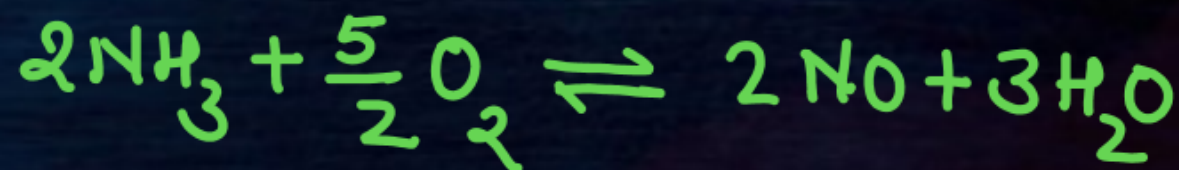
The equilibrium constant (K) of the reaction



$$\textcircled{1} + \textcircled{2} + \textcircled{3}$$

(a) $\frac{K_1 K_3^3}{K_2}$

~~(b) $\frac{K_2 K_3^3}{K_1}$~~



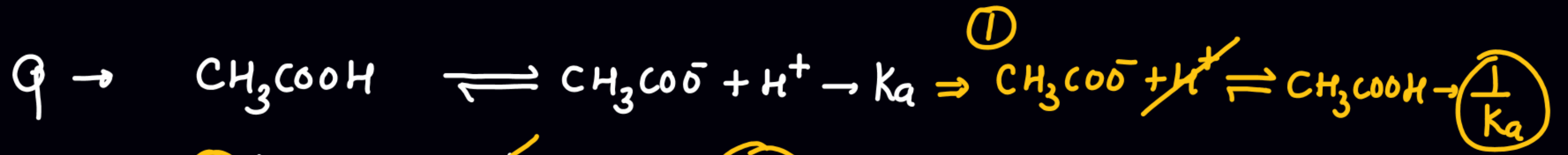
(c) $\frac{K_2 K_3}{K_1}$

(d) $\frac{K_2^3 K_3}{K_1}$

$$K = \left(\frac{1}{K_1}\right) \times K_2 \times (K_3^3)$$

$$K = \frac{K_2 \cdot K_3^3}{K_1}$$





Select correct relation

~~a) $K_h = \frac{K_w}{K_a}$~~

① + ②



b) $K_h = K_w \times K_a$

$K_h = \frac{K_w}{K_a}$

$K = \left(\frac{1}{K_a} \right) \times K_w = \frac{K_w}{K_a}$

c) $K_h = \frac{K_a}{K_w}$

d) NONE

Active Mass

Active Mass $\rightarrow$ Concentration term

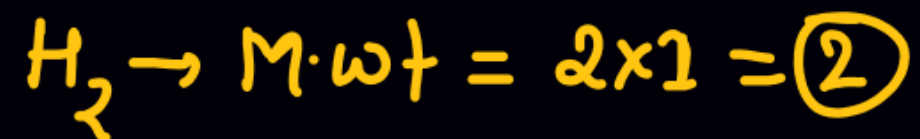
$\rightarrow$ Represent $\rightarrow$ square bracket $\rightarrow []$

$\rightarrow$ Active mass $\propto$ Molarity (M)

$$[A] = K \text{ Molarity}$$

$K=1 \rightarrow$ for aq. solⁿ, gas

$$[A] = \text{molarity} = \frac{\text{mole of comp.}}{\text{vol. of solⁿ (L)}}$$



$$[A] = \frac{\text{mole of comp.}}{\text{vol. of solⁿ (L)}}$$

Q $\rightarrow$ 4 gram H_2 react with 14 gram N_2 and form NH_3 in 10 litre close vessel.

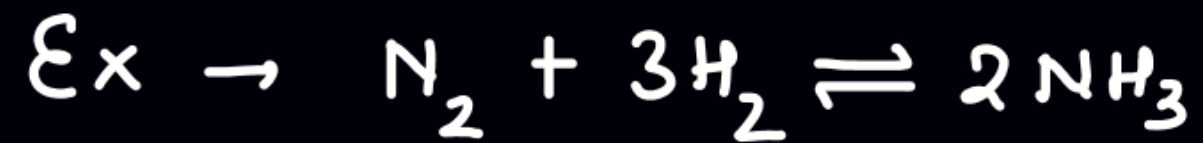
Calculate Active mass of H_2 & N_2 gas.

$$\text{solⁿ} \quad [\text{H}_2] = \frac{(\text{mole})_{\text{H}_2}}{\text{vol. of solⁿ}} = \frac{(4/2)}{10}$$

$$[\text{H}_2] = 0.2 \text{ mole/litre} = 0.2 \text{ M}$$

$$[\text{N}_2] = \frac{(14/28)}{10} = \frac{0.5}{10} = \underline{0.05 \text{ M}}$$

Eq. Constant in term of Active mass of Reactant & product

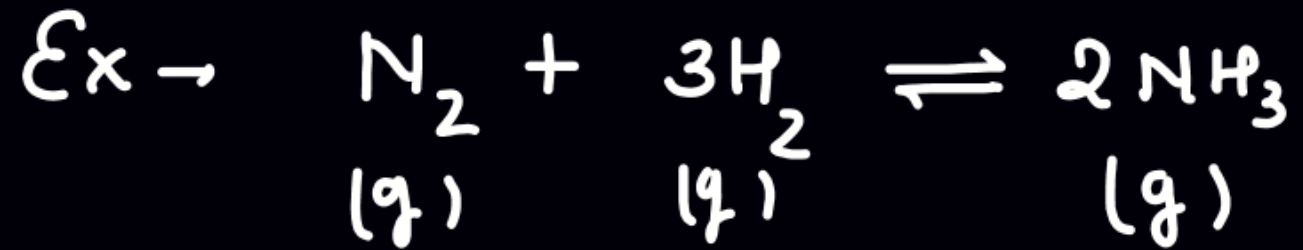



$$K_{eq} = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

$K_c \rightarrow$ Eq. Const in term of Active Mass of Reactant & prod.

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

Eq. Constant in term of Partial pressure of Reactant & product



$$\text{K}_{eq} = \frac{(P_{\text{NH}_3})^2}{(P_{\text{N}_2})^1 (P_{\text{H}_2})^3} \Rightarrow K_p = \frac{(P_{\text{NH}_3})^2}{(P_{\text{N}_2})^1 (P_{\text{H}_2})^3}$$

K_p = Eq. Const in term of
partial pressure of gas



$$P_T = P_{\text{N}_2} + P_{\text{H}_2} + P_{\text{NH}_3}$$

partial pressure = mole fractⁿ $\times P_T$

$$P_{\text{N}_2} = \chi_{\text{N}_2} \cdot P_T$$

$$P_{\text{H}_2} = \chi_{\text{H}_2} \cdot P_T$$

$$P_{\text{NH}_3} = \chi_{\text{NH}_3} \cdot P_T$$

Relation between K_p & K_c

$$K_p = K_c (RT)^{\Delta n_g} \Rightarrow \underline{RT > 1}$$

Where $\rightarrow K_p$ = Eq. Const in term of partial pressure

K_c = Eq. Const in term of Active Mass or Molar Con.

R = universal gas Const $R = 0.0821 \text{ atm} \cdot \text{litre mole}^{-1} \text{ K}^{-1}$

$$\Delta n_g = (n_g)_p - (n_g)_R$$

T = temp. of vessel

$$R = 8.314 \text{ Joule mole}^{-1} \text{ K}^{-1} \approx \frac{25}{3} \text{ J} \cdot \text{K}^{-1} \text{ mole}^{-1}$$

$$R = 1.99 \approx 2 \text{ Calori mole}^{-1} \text{ K}^{-1}$$

$$K_p = \frac{100 \times (1.2)^{-1}}{1.2} = 83.33$$

$$K_p = K_c (RT)^{\Delta n_g}$$

$$\text{or } \frac{K_p}{K_c} = (RT)^{\Delta n_g}$$

$$\Delta n_g = 0$$

$$K_p = K_c$$

$$\Delta n_g = +ve \text{ or } \Delta n_g > 0$$

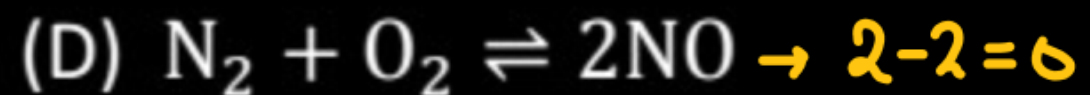
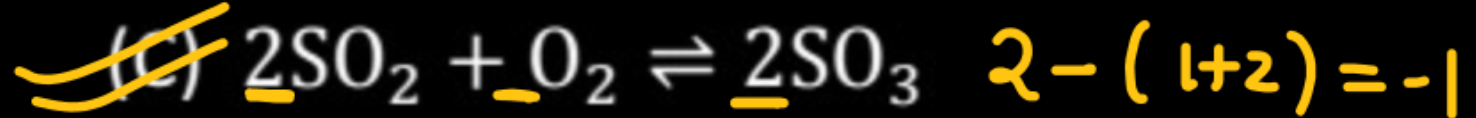
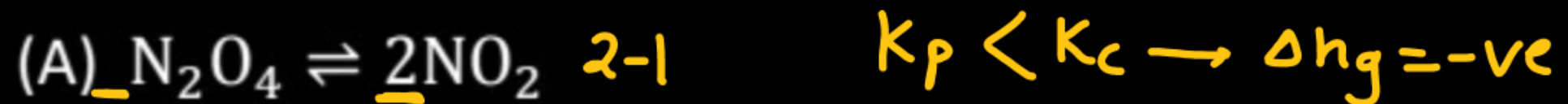
$$K_p > K_c$$

$$\Delta n_g = -ve \text{ or } \Delta n_g < 0$$

$$K_p < K_c$$



Q33. In which of the following K_p is less than K_c :



14. The ratio $\frac{K_p}{K_c}$ for reaction:



[06 April, 2024 (Shift-II)]

(a) $(RT)^{1/2}$

(b) RT

(c) 1

~~(d) $\frac{1}{\sqrt{RT}}$~~

$$\frac{K_p}{K_c} = (RT)^{\Delta n_g} = (RT)^{-1/2} = \frac{1}{(RT)^{1/2}} = \frac{1}{\sqrt{RT}}$$

$$\Delta n_g = 1 - \left(1 + \frac{1}{2}\right) = -\frac{1}{2}$$



$$\Delta n_g = 5 - 4 = 1$$

Q3. The value of K_p for the reaction $\underline{2}\text{H}_2\text{O}(\text{g}) + \underline{2}\text{Cl}_2 \rightleftharpoons \underline{4}\text{HCl}(\text{g}) + \underline{1}\text{O}_2(\text{g})$ is 0.03 atm at 427°C. The value of K_c for the same reaction is

273

~~(A) $5.22 \times 10^{-4} \text{ mol/L}$~~

(B) $7.34 \times 10^{-4} \text{ mol/L}$

(C) $3.2 \times 10^{-3} \text{ mol/L}$

(D) $5.43 \times 10^{-5} \text{ mol/L}$

$$K_p = K_c (RT)^{\Delta n_g}$$

$$0.03 \text{ atm} = K_c (0.0821 \times 700)^1$$

$$K_c = \frac{0.03}{0.0821 \times 700} = 5.22 \times 10^{-4} \text{ mole/litre}$$

For the reaction, $\text{SO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g})$, if $K_p = K_c (\text{RT})^x$ where, the symbols have usual meaning, then the value of x is (assuming ideality)

$$K_p = K_c (\text{RT})^{\Delta n_g}$$

[2014 Main]

(a) -1

$$\Delta n_g = x$$

~~(b) $-\frac{1}{2}$~~

(c) $\frac{1}{2}$

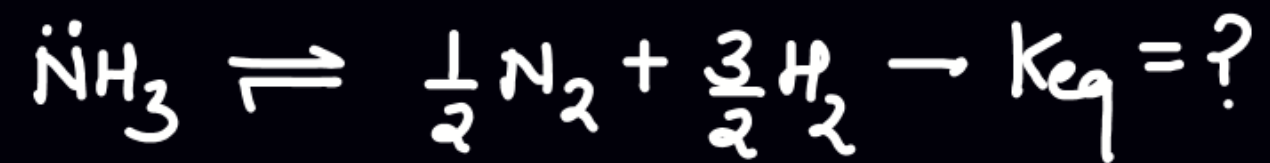
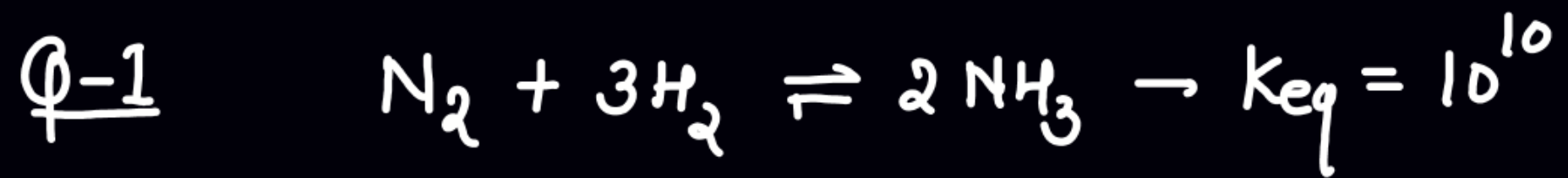
$$\Delta n_g = x = 1 - \left(1 + \frac{1}{2}\right) = -\frac{1}{2}$$

(d) 1



Chemical Equilibrium

Qsd-SHEET-2

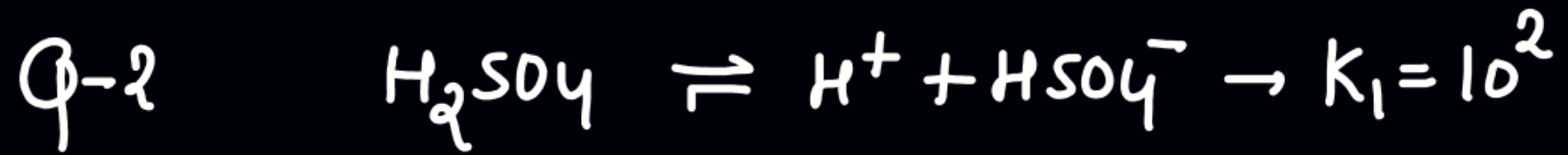


a) 10^{-5}

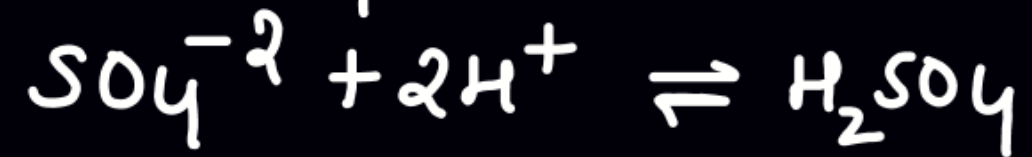
b) 10^5

c) 10^{-10}

d) 10^{-5}



Calculate Eq. Const.

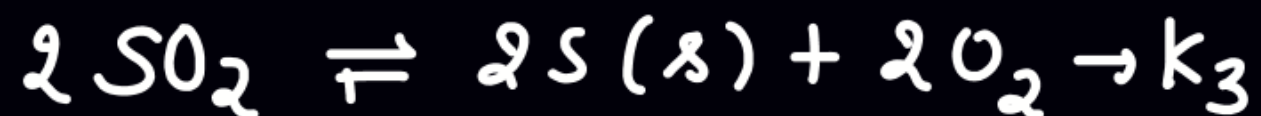
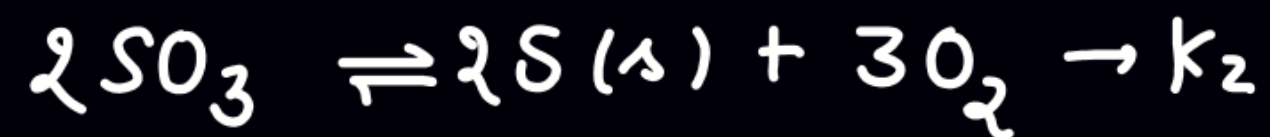
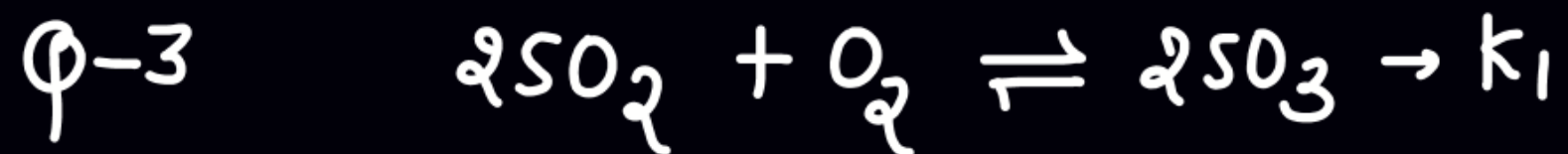


a) 10^6

b) 10^{-6}

c) 10^{10}

d) 10^{-4}



Select correct relation

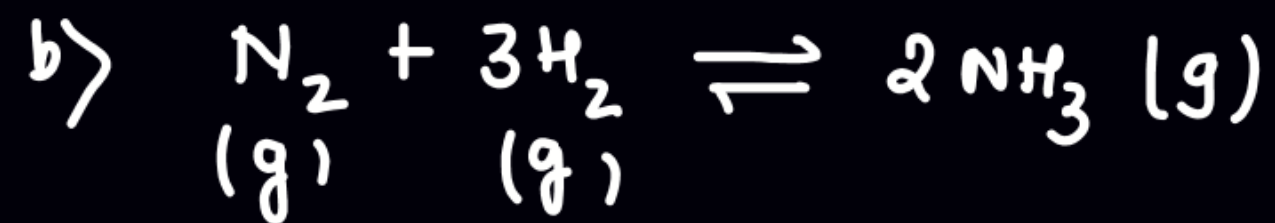
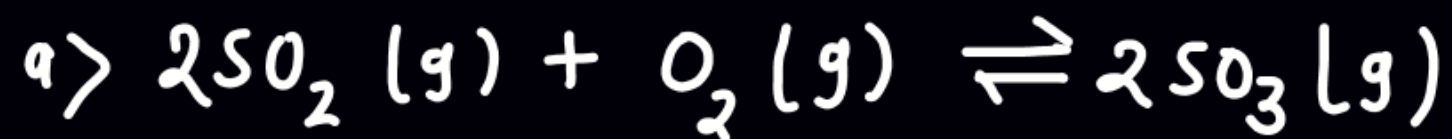
a) $K_1 = K_2 / K_3$

b) $K_1 = \frac{K_3}{K_2}$

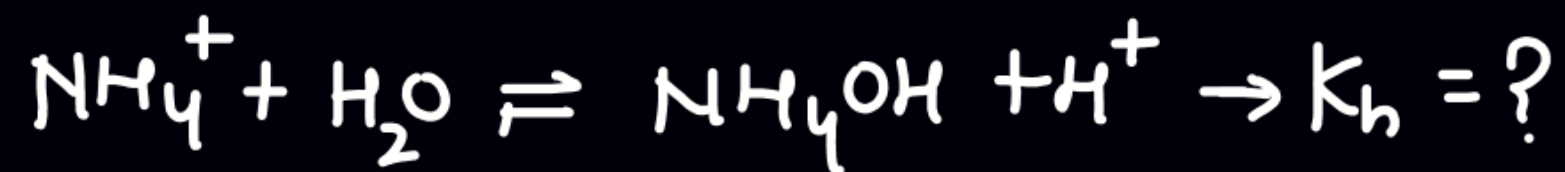
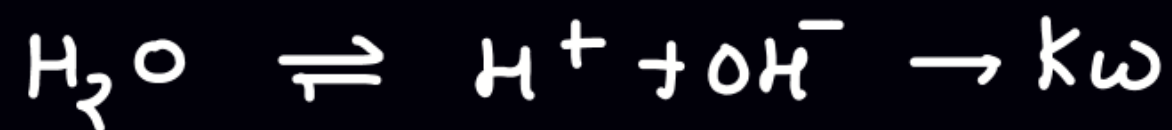
c) $K_1 = K_2 \times K_3$

d) NONE

$Q \rightarrow 4 \ln K_p = -2 \ln RT + \ln K_c$, which reaction satisfied above given Equation.



Q-5



a) $K_h = K_w \times K_b$

b) $K_h = \frac{K_w}{K_b}$

c) $K_h = \frac{K_b}{K_w}$

d) NONE

Chemical Equilibrium

Qsd-SHEET-1

Q-1 Select the correct statement at Equilibrium state.

a) It is dynamic state ✓

b) At Eq. $\Delta G = 0$ ✓

c) At Eq. $(\text{Rate})_f = (\text{Rate})_b$ ✓

d) At Eq., $\Delta G^\circ = 0$ ^f _{may be}

e) Con. of Reactant & product are constant ✓

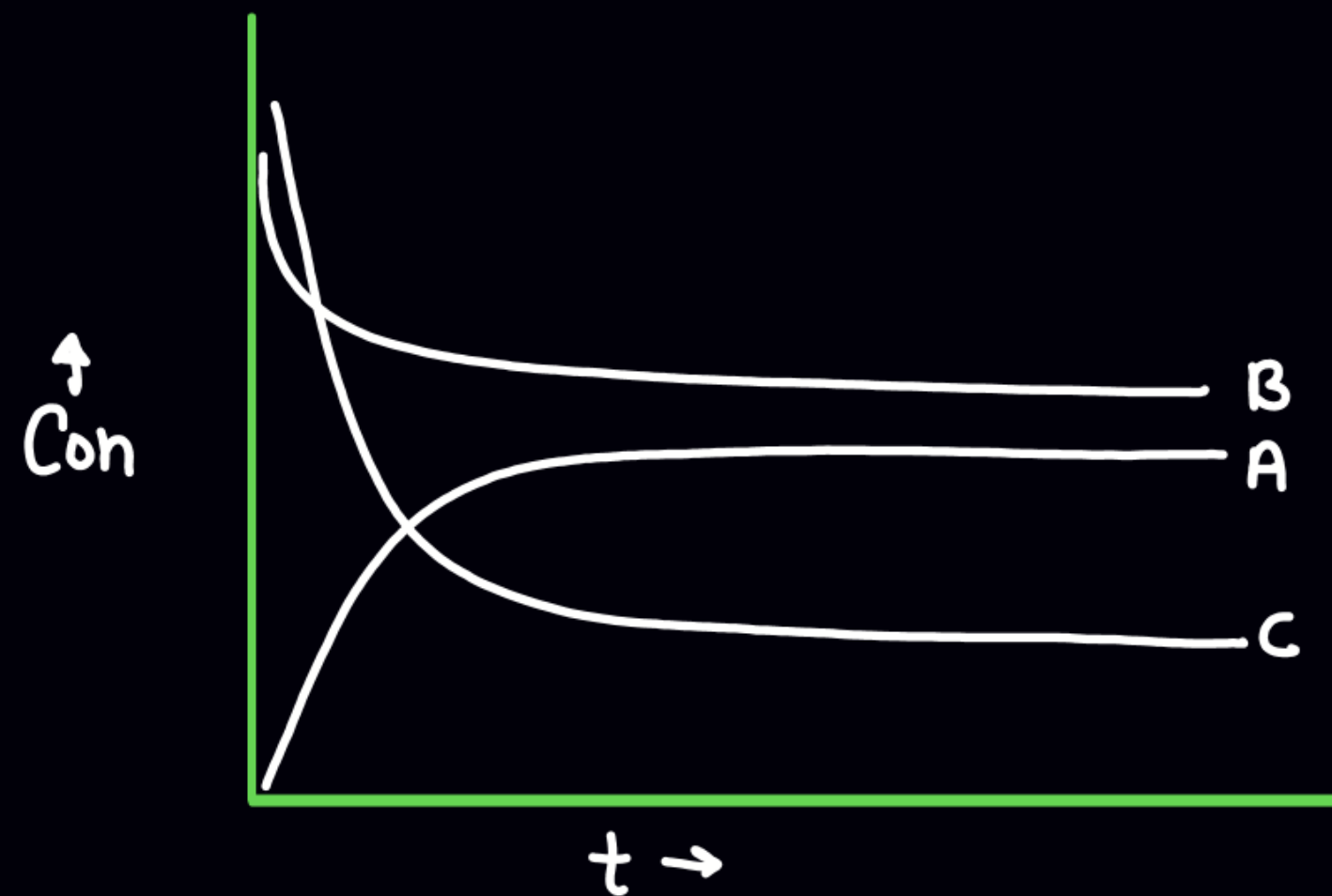
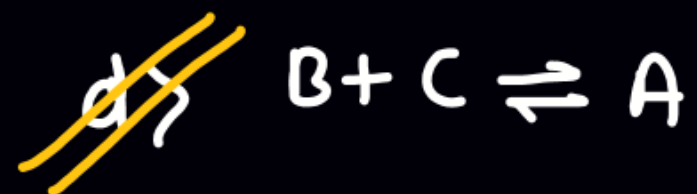
a) a, b, c, d

~~c) a, b, c, e~~

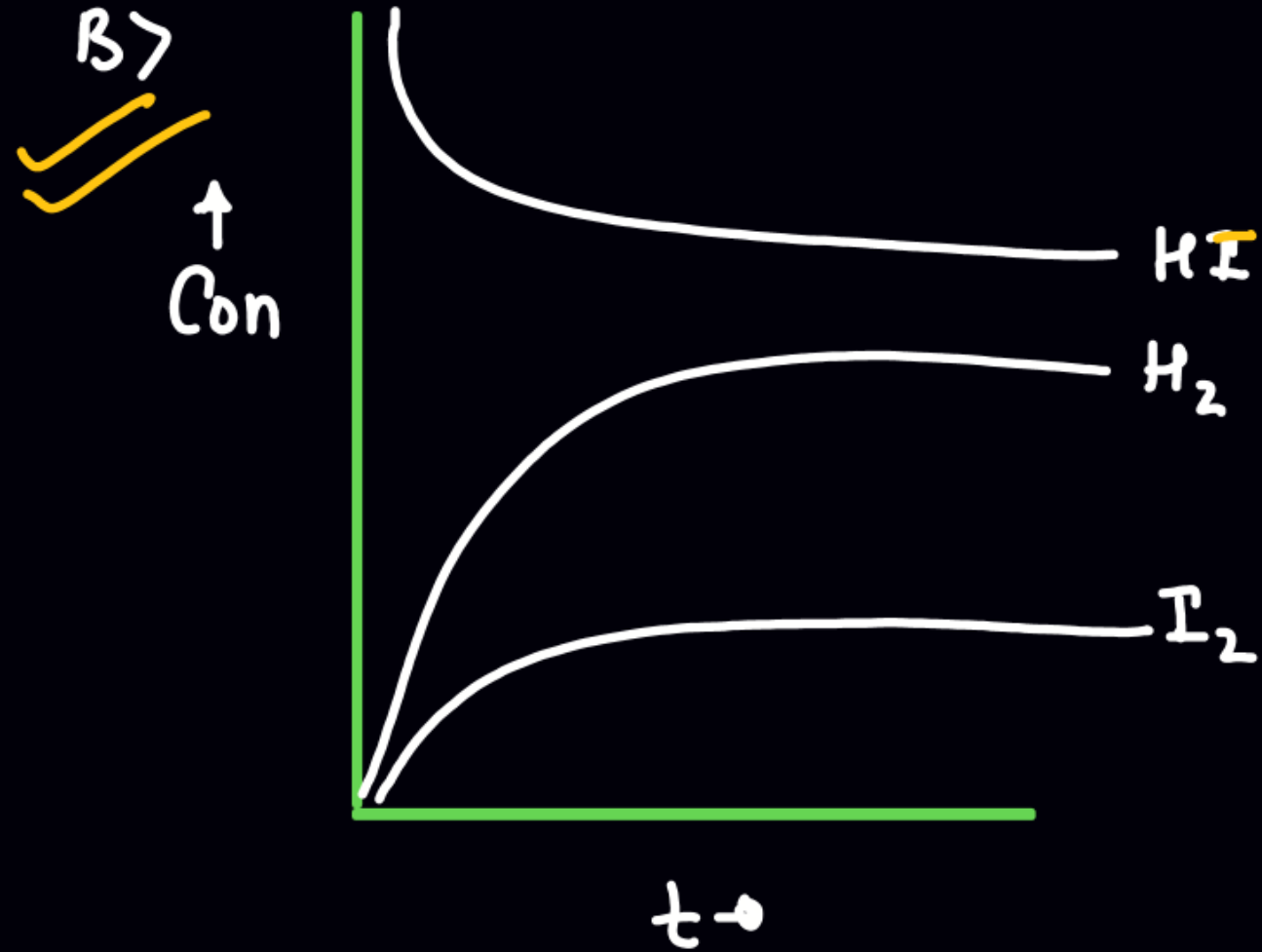
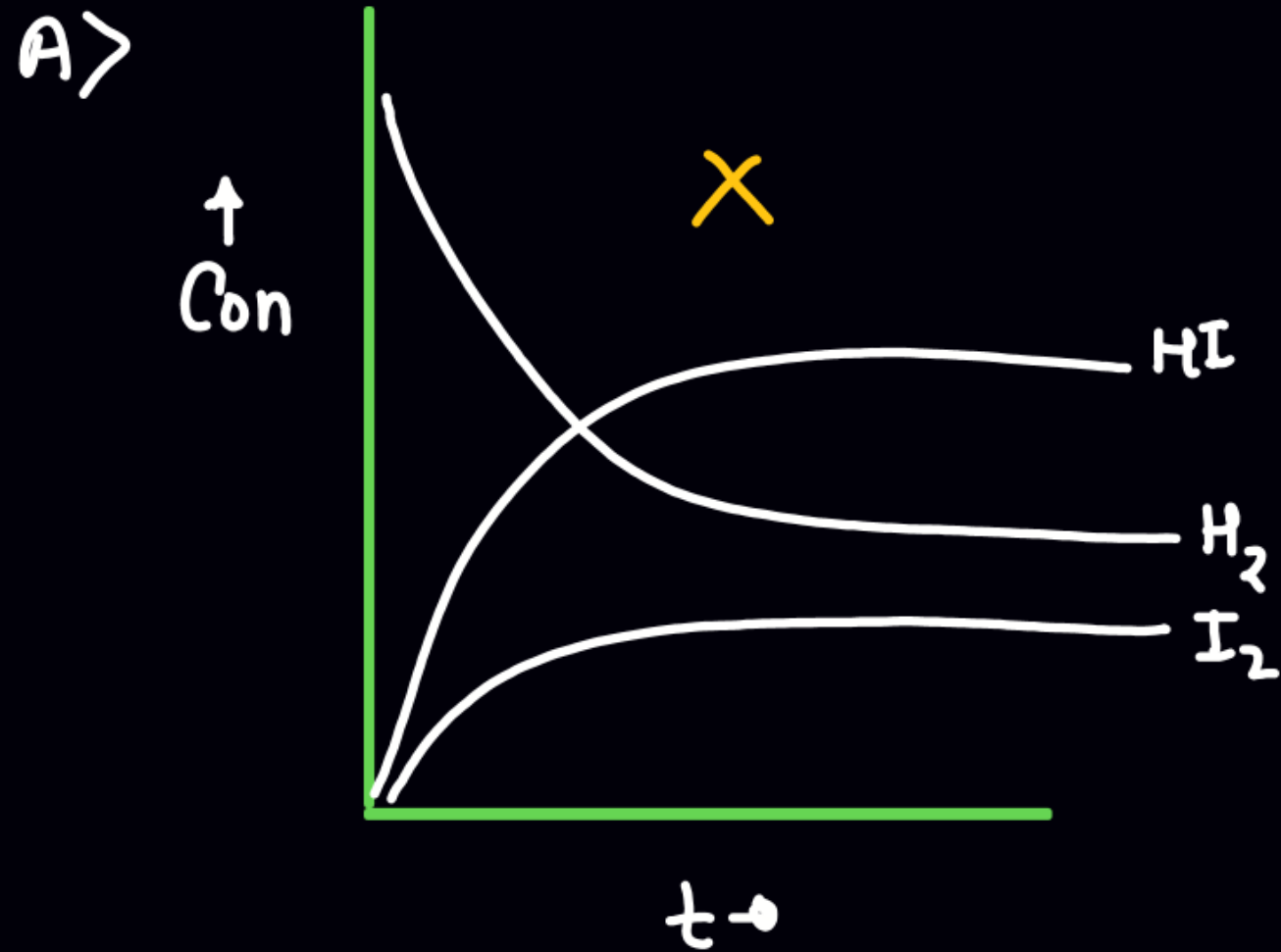
b) e, d, c

d) e, d, a

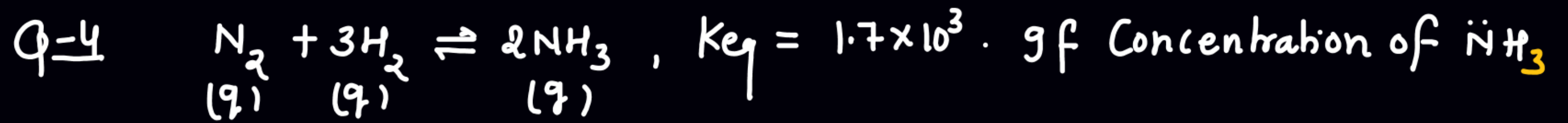
Q-2 Select Correct Equation for given graph.



Q-3. Con. of HI , H_2 & I_2 at Equilibrium 5M, 3M & 2M respectively. $2\text{HI} \rightleftharpoons \text{H}_2 + \text{I}_2$. Select Correct graph.



C> NONE



increases 10 times then Value of Eq. Constant is

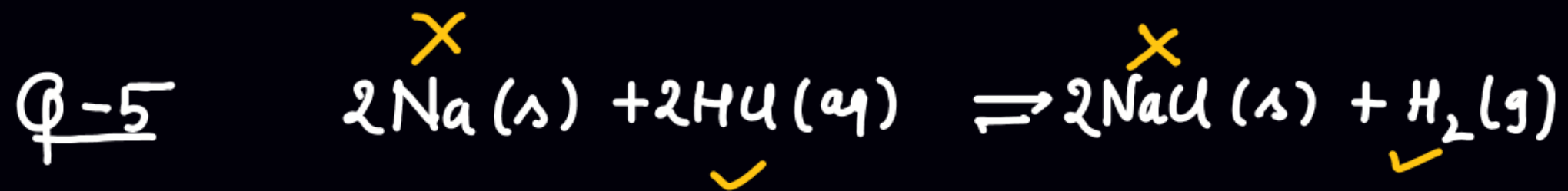
a) increases 100 times

b) increases 10 times

c) decreases $\frac{1}{100}$ times

~~d) Remains same~~

$K_{eq} \rightarrow$ only depend upon temp.



K_{eq} is.

a) $K_{eq} = \frac{[\text{NaCl}]^2 [\text{H}_2]}{[\text{Na}]^2 [\text{HCl}]^2}$

b) $K_{eq} = \frac{[\text{NaCl}]^2 [\text{H}_2]}{[\text{Na}] [\text{HCl}]^2}$

c) $K_{eq} = \frac{[\text{NaCl}]^2 [\text{H}_2]}{[\text{HCl}]^2}$

~~d) $K_{eq} = \frac{[\text{H}_2]}{[\text{HCl}]^2}$~~

$K_{eq} = \frac{[\text{H}_2]}{[\text{HCl}]^2}$